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MATERIAL AND METHODS

Charts of patients with DMD followed by our neuro-orthopedics outpatients' clinics were retrospectively reviewed. This study was approved by the Hospital's Ethics Committee (CAAE 29523920.0.0000.0068).

Population

The study was conducted at the Orthopedics Department of a tertiary hospital. All patients submitted to surgery or on a waiting list for surgery during the period from January, 2012 to December, 2020 were analyzed for inclusion, thus defining a convenience sample. Inclusion criteria were: confirmed diagnosis of DMD (either by genome or muscle biopsy); minimum 24 months of post-operative follow-up; complete chart, containing data concerning the time of gait interruption, date of surgery, description of the performed surgical procedure, retainment or not of gait ability after surgery, and occurrence of postoperative complication. In cases where the chart was incomplete, the patient's family or caregiver was interviewed.

Exclusion criteria consisted of: muscular dystrophies other than Duchenne's; incomplete chart or unreliable information; post-operative follow-up shorter than 24 months. Age was not a criterion for inclusion or exclusion.

Included patients were divided into two groups: patients in group 1 had stopped walking before surgery, and patients in group 2 were still walking before surgery. If present, information on medication in use was included.

At our institution, surgery is recommended for all DMD patients who present tendon contractures, regardless of the patient's ambulatory status. Surgery might be indicated even for wheelchair bound patients, for comfort and/or hygienical purposes, according to parents' or legal guardians' desire.

The Covid-19 pandemic had an important impact in our services, and elective surgical procedures were interrupted for over 15 months, from March, 2020 on.

Statistical Analysis

In this study, a good outcome after surgery was considered when the patient retained or recovered the ability to walk. Three clinical questions were addressed:

1. Is it necessary to intervene before losing gait?
2. Is there a relationship between age and a good surgical outcome?
3. Is it more effective to perform the orthopedic procedure less than 12 months before the patient stopped walking?

Based on our limited sample size, we approached the first question using the Monte Carlo Sampling Algorithm,¹⁵ then we applied a Fisher Exact Test.¹⁶

To answer the second question, we applied a T-Test¹⁷ and performed a logistic regression analysis.¹⁸

For the third question, we applied a Hypothesis Test, a T-Test, and a Logistic Regression.

RESULTS

A total of 49 patients were identified. Their outcomes and distribution are presented in Figure 1 and Tables 1 and 2.

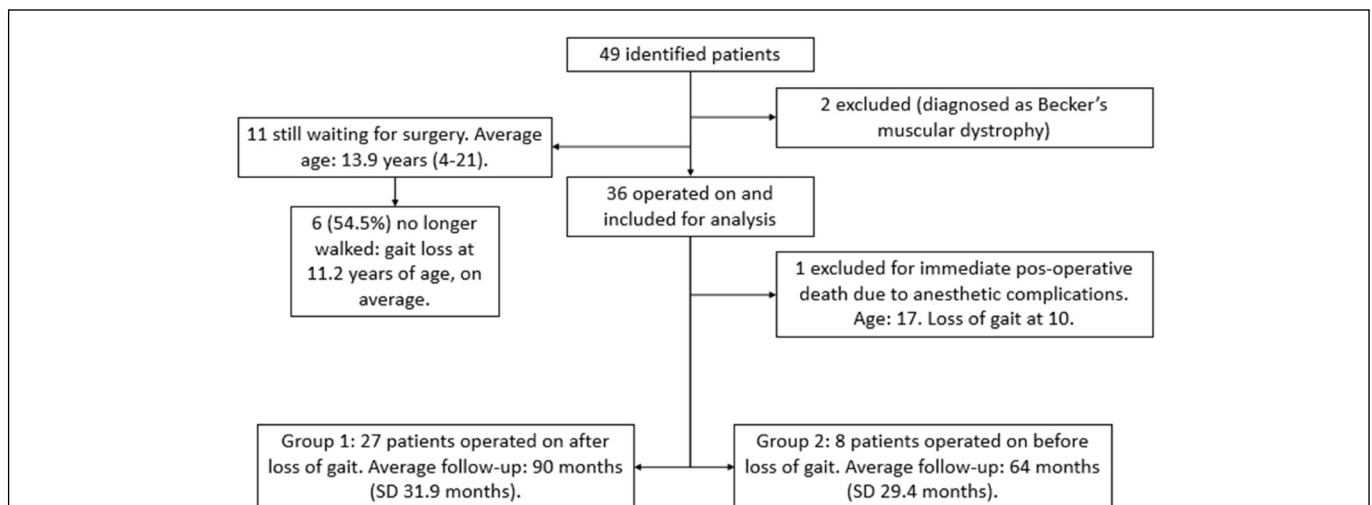
Single-level surgeries consisted of either uni- or bilateral Achilles' tendon lengthening. Multi-level surgeries consisted of unilateral or bilateral tenotomies at the hips (iliopsoas, fascia lata, gluteus), knees (ischiotibial muscles, femoral biceps), and ankles (Achilles' tendon), and were performed as required for adequate contracture release.

Timing of surgery in relation to gait loss and its effects

Groups 1 and 2 were compared for this analysis. There was a statistically significant difference concerning surgical results (gait maintenance/recovery) between both groups, in favor of the group that was still walking at the moment of surgery (29.6% recovered gait in Group 1, and 75% maintained gait in Group 2, $p = 0.04$).

Age at the procedure and gait recovery after the procedure

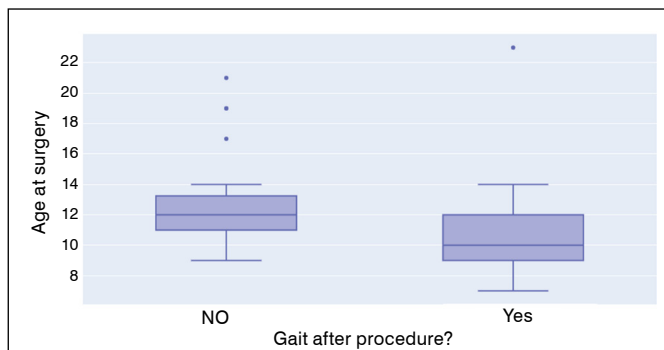
Within group 1, patients who recovered gait after the procedures were significantly younger than those who did not (average 10 years vs 12.9 years, $p = 0.$), as shown in Figure 2. Logistic regression suggested that the younger the patient was, the higher his chance of recovering gait, increasing by 5% per year younger (considering the sample age interval of 5 to 23 years old).



		(in months)				(in months)	Not available
		120	19	M	No	NA	Yes
		11	11	M	No	NA	Yes
		36	10	M	No	NA	NA
4	7	36	10	M	No	NA	NA
5	9	120	9	M	No	NA	Yes
6	12	5	12	S	No	NA	Yes
7	11	12	12	M	No	NA	Yes
8	12	12	13	M	No	NA	NA
9	11	24	13	M	No	NA	NA
10	10	7	11	M	No	NA	Yes
11	17	24	19	M	No	NA	NA
12	11	9	12	M	No	NA	NA
13	10	24	11	M	No	NA	Yes
14	8	12	9	M	Yes	3	Yes
15	10	12	11	M	No	NA	NA
16	13	7	13	S	No	NA	NA
17	11	7	11	M	No	NA	NA
18	12	108	21	M	No	NA	Yes
19	10	17	11	M	No	NA	Yes
20	12	60	17	M	No	NA	Yes
21	10	24	12	M	Yes	120	Yes
22	13	12	14	M	Yes	30	Yes
23	8	2	8	M	Yes	30	NA
24	10	8	11	M	Yes	42	Yes
25	7	6	8	M	Yes	7	NA
26	8	9	9	M	Yes	Retains gait	NA
27	9	7	9	S	Yes	Retains gait	Yes

Table 2. DMD patients operated on before gait loss (G2).

Patient number	Age at surgery (in years)	Single (S) or Multi (M) level surgery	Gait maintained after procedure? (Yes/No)	Interval until new disability (in months)	Glucocorticoids use (Yes/No/NA = Not available)
1	7	M	Yes	Retains gait	Yes
2	11	S	Yes	Retains gait	NA
3	12	S	Yes	Retains gait	Yes
4	23	M	Yes	Retains gait	Yes
5	10	M	No	NA	NA
6	12	M	Yes	11	Yes
7	9	M	Yes	3,5	Yes
8	14	S	No	NA	Yes



Interval between gait loss and surgery

Patients from group 1 were evaluated concerning the interval, in months, between gait loss and surgical procedure, as well as the surgical outcome (Figure 3). Those who recovered gait after the procedure had a shorter interval than those who did not recover gait (10 vs 34.26 months, $p = 0.001$).

Time limit for intervention after the loss of gait

Different intervals were tested (3, 6, and 12 months) as the defining limit for the loss of gait. Procedures performed before 12 months of gait loss presented statistically significant better results than procedures performed after 12 months ($p = 0.014$). By logistic

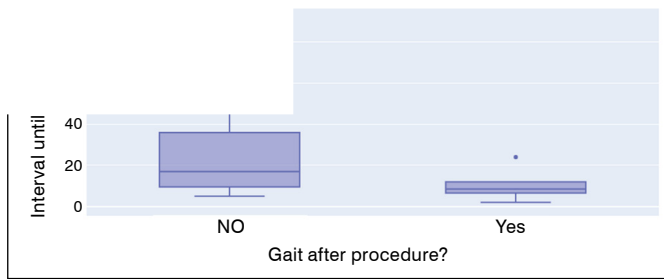


Figure 3. Distribution of the interval between loss of gait and surgery between in group 1 and the outcome (gait recovery).

in average; and each day of delay represented a 0.2% decrease in chances to recover gait.

When comparing different intervals (Figure 4), it was possible to observe that chances of success were higher in the first six months and each month of delay represented an 11% decrease in success rate. In the following six months, the decrease of chance of success was less intense (from 6 to less than 12 months, each month of delay represented a 7% decrease), and, finally, it was the smallest in the last interval (4% for each month of delay).

Gait recovery or maintenance depends on the patient's inherent chance of success, which is directly related to age and gait status. That is, chances of success are even slimmer for the older, non-ambulatory patients who wait longer for surgery.

Surgical procedure technique: single- or multi-level

Groups 1 and 2 were evaluated both separately and combined. In total, of those patients who underwent multi-level surgery, 10 recovered gait, and 18 did not (64,3%). Of those who underwent single-level surgery, four recovered gait, and two did not (33%) ($p=0.34$).

Glucocorticoid use

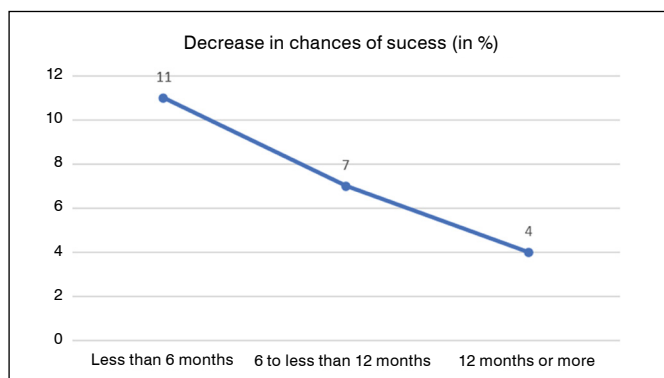
This information was missing in 14 patients' charts and remained unknown even after direct questioning, therefore it was impossible to analyze GC's influence on gait. However, all patients with available information (21 out of 35) used GC, what is in accordance with our institutional protocol.

Average extended time of gait after surgery

For those patients who recovered gait after surgery, the average extended time of gait was of 38.6 months before a new disability, which occurred at 14.9 years of age, on average. Of the eight patients operated on before losing gait, two ceased walking after surgery,

from both groups who were still walking at the end of follow-up.

DISCUSSION



as the first (and only) one to present d of a retrospective analysis of cases 2009. Out of the 17 DMD cases op- of 12.4 years, three were still walking before surgery and presented a gait prolonged for 8 more months (6-12 months), on average. The other 14 patients were unable to walk at the moment of surgery and presented no recovery afterwards, no matter if the loss was "a little before" or "much before" the surgery (the exact period of time was not described). Surgical technique and indication were very similar to ours. The authors conclude that procedures should be indicated when the patient is about to lose gait, with the intention to recover standing ability, more than recovering the ability to ambulate. Their results, as ours, once again highlight the role of age, when considering surgical treatment for DMD.

Despite so much evidence, the current consensus on the treatment of DMD¹⁴ affirms that surgical treatment should only be indicated during the walking phase, and exclusively for Achilles' tendon lengthening, if it compromises gait. This consensus, however, is based on specialists' opinions and disregards several decades of evidence in favor of surgical treatment in DMD, as shown above. Our findings, along with previous studies, indicate that patients with DMD may benefit from surgical procedures of multilevel lengthening

months of time after gait loss – and, by logistic regression, estimates how much each month of delay compromises surgical results.

Study limitations

This study was a retrospective cohort; therefore, there is a limitation in the data that can be obtained in group comparison. Even though the lack of data concerning GC does not compromise the results, it decreases the possibilities of analysis between different groups. The sample size was small, which limits the strength of conclusions that can be drawn, particularly in subgroup analyses. Nevertheless, the sample size was considerable when compared to similar studies. It is representative of the population that is treated in our outpatient clinic; however, more studies are necessary (ideally, clinical trials) to help in defining new guidelines to optimize outcomes for this patient cohort.

CONCLUSION

Surgical treatment for DMD extended gait for 38.6 months, on average in our patient cohort. The sooner it is performed, the better the results. The outcomes for single or multilevel surgeries did not show statistical differences. Surgery may be performed after the loss of gait and still have positive results, however, the time limit for intervention is less than 12 months. Each day of delay after gait loss decreases the chances of success by 0.2%.

AUTHOR'S CONTRIBUTION: Each author made significant individual contributions to the development of this manuscript. DN, PMG, AP, CS, MBDR: Study Design; DV, FR: Data collection; DN, PMG: Data interpretation; DN, PMG: Drafting Manuscript.

REFERENCES